

Thermal Engineering

ME 1504

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Course

SEMESTER V

Code No.	Subject	L Hrs/ Wk	T Hrs/ Wk	P Hrs/ Wk	C	Int.	Univ.	Total
CE/CS/EB/E C/EE/EI/FT/ IT/ME/ SE 1501	Engineering Mathematics IV	3	1	0	3	50	100	150
ME 1502	Mechanics of Machinery	3	1	0	3	50	100	150
ME 1503	Machine Tools & Machining Science	3	1	0	3	50	100	150
ME 1504	Thermal Engineering	3	1	0	3	50	100	150
ME 1505	Industrial Management	3	1	0	3	50	100	150
ME 1506	Power Plant Engineering	3	1	0	3	50	100	150
ME 15 L1	Computational Methods Lab	0	0	3	2	100	-	100
ME 15 L2	Machine Shop	0	0	3	2	100	-	100
	TOTAL	18	6	6	22			

Course

ME 1504 : THERMAL ENGINEERING

Module I

Fuels and combustion – Solid, liquid and gaseous fuels – calorific value – calorimeter – combustion equation – Air – Fuel ratio gravimetric & volumetric analysis – excess air Enthalpy and Internal Energy of Combustion – application of first law of thermodynamics to chemical reaction (combustion), adiabatic flame temperature – application of second law of thermodynamics to chemical reaction.

Module II

Air standard cycles, Otto, Diesel, Dual, Brayton, Stirling cycles. Actual cycles of four stroke and two stroke IC Engines, valve timing diagram – Engine testing – Performance and characteristics of constant speed and variable speed engines – heat balance test – Morse test – retardation test – effect of dissociation – variable specific heats and heat losses – scavenging – objectives – effects and methods – Efficiencies (thermal, mechanical and volumetric efficiencies)

Module III

Systems and components of IC Engines – fuel systems – Ignition systems – Cooling – starting – lubrication – governing of IC engines – super charging of SI and CI Engines – turbo charging – exhaust emissions of IC engines – alternate Potential Engines – free piston engines – Wankel Engine and Stratified charged engine automotive transmission system and its components.

Module IV

Combustion in IC engines – flame propagation normal and abnormal combustion detonation – Pre ignition – after burning – HUCR – fuel rating – additives in petrol – combustion chambers of SI engines – combustion in CI engines – phase of normal combustion diesel knock – effect of engine variables on diesel knock – cetane number – additives in diesel – combustion chambers of CI engines.

References:

- 1) Cengel and Boles, Thermodynamics- An Engineering Approach, Tata McGraw Hill
- 2) M.I. Mallev, Internal Combustion Engines
- 3) Engineering Thermodynamics Bacon, Butterworth
- 4) Elements of Internal Combustion Engine Rogowsky, Tata McGraw Hill
- 5) Fundamentals of Internal Combustion Engines Gill, Smith & Ziurys, Oxford
- 6) Modern Petrol Engine Judge, Chapman & Hall
- 7) Applied Thermodynamics for Engineering Technology sts- T.D Eastop , A Mc Conkey Pearson Education

Type of Questions for University Exam.

Q 1.Eight short answer questions of 5 marks with two questions from each of the four modules. (8x5 = 40 marks)

Q 2. to Q.5 : Two questions A & B of 15 marks from each modules with option to answer either A or B. (4x15 = 60

Course Outcomes- ME 1504 Thermal Engineering.

At the end of the course, the student will be able to:

CO-1 To understand fuels, calorimetry, basics of combustion.

CO-2 Applying thermodynamic laws to combustion.

CO-3 To understand the air standard cycles used in various thermal equipments, its thermodynamic analysis.

CO-4 Actual working of Internal Combustion Engines, its performance analysis.

CO-5 To understand various systems in Internal Combustion Engines components and its working.

CO-6 Alternate potential engines working.

CO-7. Understanding the combustion in Internal Combustion Engines.

CO-8 Dealing with abnormal combustion in Internal Combustion Engines.

Units, System International

- 1 Length - m
- 2 Mass - kg
- 3 Time -s
- 4 Temperature - K
- 5 Force - N, $1N = 1kgm/s^2$
- 6 Energy - J
- 7 Power - rate of energy transfer or storage - W
=J/s, $1kW = 1000W$

- 1 Pressure - Pa,
 $1Pa = 1N/m^2$
- 2 $1\text{ bar} = 10^5\text{ Pa} = 100\text{ kPa},$
 $= 0.1\text{ Mpa}$
- 3 $1\text{ atm} = 101.325\text{ kPa}$
- 4 Gauge pressure, Absolute pressure.
- 5 Density kg/m^3
- 6 Specific volume m^3/kg

Fuels

- 1 Fuels - A fuel is any material that can be made to react so that it releases chemical or nuclear energy as heat to be used for work
- 2 Fossil fuels - fuels formed by natural processes such as anaerobic decomposition of buried dead organisms, containing energy
- 3 Solid fuel
 - wood, charcoal, peat, coal
- 4 Liquid fuel
 - Petroleums (Gasoline, Diesel, etc.), Bio Diesel, Ethanol, Methanol
- 5 Gaseous fuel
 - Natural Gas (Methane), Liquefied petroleum gas (mixture of propane and butane)

Coal



Calorific Value

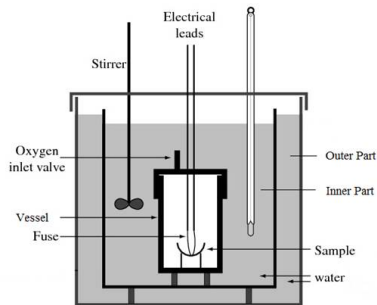
- The heat of combustion is the total energy released as heat when a substance undergoes complete combustion with oxygen under standard conditions ,i.e. to 0 °C and under a pressure of 1 bar. Unit kJ/kg
- Higher Calorific Value (or Gross Calorific Value - GCV, or Higher Heating Value - HHV) - the water of combustion is entirely condensed and that the heat contained in the water vapor is recovered.
- Lower Calorific Value (or Net Calorific Value - NCV, or Lower Heating Value - LHV) - the products of combustion contains the water vapor and that the heat in the water vapor is not recovered
- Diesel HCV = 47000 kJ/kg LCV = 43000 kJ/kg
- Petrol HCV = 45700 kJ/kg LCV = 42900 kJ/kg

Calorimeter

Bomb calorimeter

- 1 A calorimeter is an equipment used for calorimetry, or the process of measuring the heat of chemical reactions
- 2 A bomb calorimeter is a type of constant-volume calorimeter
- 3 Used to find CV of solid and liquid fuels
- 4 Heat of combustion of any compounds by

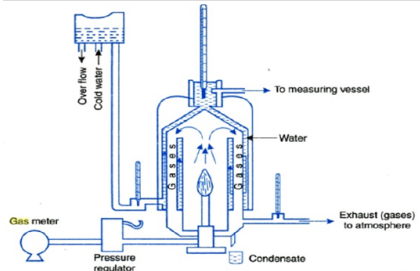
$$C.V. = [C_{(steel+water)}\Delta T]/m_{sample}$$



Junkers Calorimeter

Junkers calorimeter

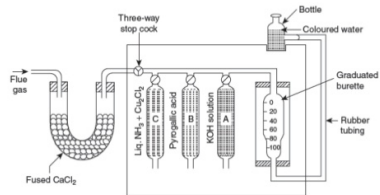
- 1 Used to find CV of gaseous fuels
- 2 burning of a known volume of gas
- 3 imparting the heat with maximum efficiency to steadily flowing water
- 4 $\text{HCV of the fuel} = \frac{wC_p(T_2 - T_1)}{V}$,
- 5 w - wt of water passed in time t
- 6 V = volume of gas burnt at S.T.P in time t
- 7 T_1, T_2 Temperature of water coming in, out.



Orsat apparatus

- 1 Known volume of flue gas admitted
- 2 CuCl_2 removes Carbon monoxide
- 3 KOH removes Carbon dioxide
- 4 Pyrogallor removes oxygen
- 5 % reduction in volume indicates the respective gas volume present in flue gas

Orsat's apparatus

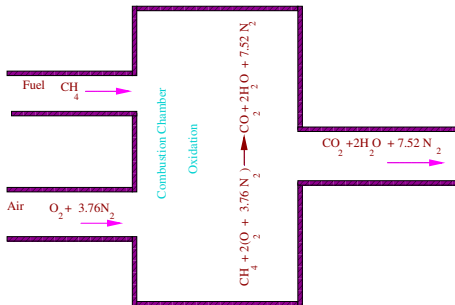


Combustion

Combustion

- 1 Oxidation of constituents in the fuel
- 2 Oxygen from atmospheric air
- 3 21 % Oxygen and 79 % Nitrogen by Volume
- 4 $\text{Air} \Rightarrow \text{O}_2 + 3.76\text{N}_2$
- 5 Minimum amount of air supplying sufficient oxygen for the complete combustion of fuel constituents- Stoichiometric air (Theoretical air)

Combustion chamber



Volumetric analysis

Volumetric analysis of gaseous mixture

Constituent	% by Volume	Mole Fraction	Molecular weight	Mass kg/kmol	Mass basis %
CO ₂	12	0.12	× 44	5.28	17.55
O ₂	4	0.04	× 32	1.28	4.26
N ₂	82	0.82	× 28	22.96	76.33
CO	2	0.02	× 28	0.56	1.86

Gravimetric analysis

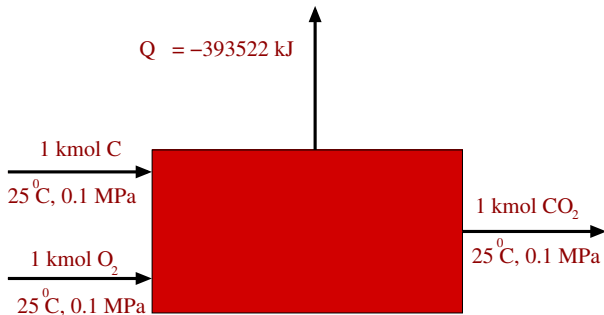
Gravimetric analysis of a mixture

Constituent	Mass Fraction	Molecular weight	kmol/kg	Mole fraction	Volumetric Analysis %
CO ₂	0.1755	÷ 44	0.00399	0.12	12
O ₂	0.0426	÷ 32	0.00133	0.04	4
N ₂	0.7633	÷ 28	0.02726	0.82	82
CO	0.0186	÷ 28	0.00066	0.02	2
			0.03324	1.0	100

Combustion of fuel

- Combustion of Octane - C_8H_{18}
- $C_8H_{18} + 12.5O_2 + 12.5(3.76)N_2 \implies 8CO_2 + 9H_2O + 47N_2$
- Air fuel ratio = $\frac{\text{mass-of-air}}{\text{mass-of-fuel}}$
- Equivalence ratio $\phi = \frac{\text{Actual-fuel-air-ratio}}{\text{Stoichiometric-fuel-air-ratio}}$,
 $\phi=1$ - Stoichiometric-fuel-air-ratio,
 $\phi<1$ -lean mixture, $\phi>1$ -rich mixture.
- Air fuel ratio on a mole basis = $AF = \frac{12.5+47}{1} = 59.5$ mol
air/mol fuel
- Air fuel ratio on a mass basis = $AF = \frac{59.5 \times 28.97}{114.2} = 15.0$ kg
air/ kg fuel.
- Excess air - % more air than the Stoichiometric ratio,
- 15 % excess air $\implies \frac{115}{100} \times$ Stoichiometric ratio

Enthalpy of formation



Chemical reaction

- 1 Reactants and products are at 25 °C and 0.1 MPa
- 2 Heat transfer from control volume -393522 kJ.

Enthalpy of formation

- At the reference state enthalpy of elements are assumed to be Zero
- Considering the above reaction $C + O_2 \rightarrow CO_2$
- Applying the first law to the reaction $Q_{c.v} + H_R = H_P$
- General form $Q_{c.v} + \sum_R n_i h_i = \sum_P n_e h_e$
- For the reaction $Q_{c.v} = H_P = -393522 \text{ kJ/kmol}$
- Enthalpy of formation for CO_2 $h_f^0 = -393522 \text{ kJ/kmol}$
- Enthalpy at any pressure and Temperature $h_{T,P} = h_f^0 + \Delta h_{T,P}$

Enthalpy of formation

TABLE A-30 Enthalpy of Formation, Gibbs Function of Formation, and Absolute Entropy of Various Substances at 298 K and 1 atm

$\bar{h}_f^\circ$ and $\bar{g}_f^\circ$ (kJ/kmol), $\bar{s}^\circ$ (kJ/kmol · K)

Substance	Formula	$\bar{h}_f^\circ$	$\bar{g}_f^\circ$	$\bar{s}^\circ$
Carbon	C(s)	0	0	5.74
Hydrogen	H ₂ (g)	0	0	130.57
Nitrogen	N ₂ (g)	0	0	191.50
Oxygen	O ₂ (g)	0	0	205.03
Carbon monoxide	CO(g)	-110,530	-137,150	197.54
Carbon dioxide	CO ₂ (g)	-393,520	-394,380	213.69
Water	H ₂ O(g)	-241,820	-228,590	188.72
Water	H ₂ O(l)	-285,830	-237,180	69.95
Hydrogen peroxide	H ₂ O ₂ (g)	-136,310	-105,600	232.63
Ammonia	NH ₃ (g)	-46,190	-16,590	192.33
Oxygen	O(g)	249,170	231,770	160.95
Hydrogen	H(g)	218,000	203,290	114.61
Nitrogen	N(g)	472,680	455,510	153.19
Hydroxyl	OH(g)	39,460	34,280	183.75
Methane	CH ₄ (g)	-74,850	-50,790	186.16
Acetylene	C ₂ H ₂ (g)	226,730	209,170	200.85
Ethylene	C ₂ H ₄ (g)	52,280	68,120	219.83
Ethane	C ₂ H ₆ (g)	-84,680	-32,890	229.49
Propylene	C ₃ H ₆ (g)	20,410	62,720	266.94
Propane	C ₃ H ₈ (g)	-103,850	-23,490	269.91
Butane	C ₄ H ₁₀ (g)	-126,150	-15,710	310.03
Pentane	C ₅ H ₁₂ (g)	-146,440	-8,200	348.40
Octane	C ₈ H ₁₈ (g)	-208,450	17,320	463.67
Octane	C ₈ H ₁₈ (l)	-249,910	6,610	360.79
Benzene	C ₆ H ₆ (g)	82,930	129,660	269.20
Methyl alcohol	CH ₃ OH(g)	-200,890	-162,140	239.70
Methyl alcohol	CH ₃ OH(l)	-238,810	-166,290	126.80
Ethyl alcohol	C ₂ H ₅ OH(g)	-235,310	-168,570	282.59
Ethyl alcohol	C ₂ H ₅ OH(l)	-277,690	174,890	160.70

Sources: Adapted from K. Wark, *Thermodynamics*, 4th ed., McGraw-Hill, New York, 1983, as based on JANAF Thermochemical Tables, NSRDS-NBS-37, 1971; *Selected Values of Chemical Thermodynamic Properties*, NBS Tech. Note 270-3, 1968; and *API Research Project 44*, Carnegie Press, 1953.

Chemical reaction

- 1 Enthalpy of formation found using statistical thermodynamics
- 2 Reference state 25 °C, 0.1 MPa
- 3 Same substance may exist in different forms , one state given the reference status and other conditions defined relative to this

First law analysis of Reacting Systems

Chemical reaction

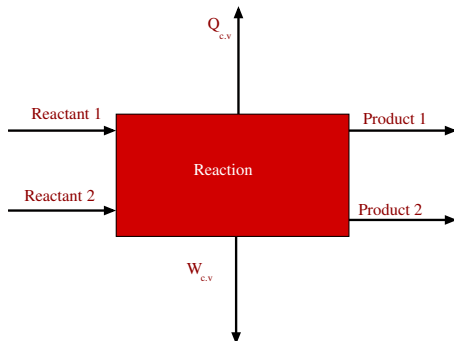
- 1 Reactants and products enthalpies available at 25 °C and 0.1 MPa
- 2 The enthalpies can be found for any different working conditions

- 3 Steady flow steady state first law $Q_{c.v} + H_R = W_{c.v} + H_P$

- 4 In general

$$Q_{c.v} + \sum_R n_i h_i = W_{c.v} + \sum_P n_e h_e$$

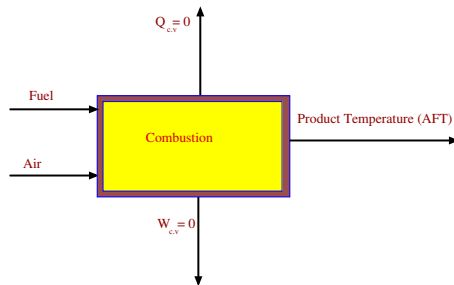
- 5 $Q_{c.v} + \sum_R n_i [h_f^0 + \Delta h]_i = W_{c.v} + \sum_P n_e [h_f^0 + \Delta h]_e$



Adiabatic Flame Temperature

Adiabatic Combustion

- 1 No work and heat transfer across boundary
- 2 No change in K.E and P.E
- 3 Maximum adiabatic flame temperature achievable with Stoichiometric ratio.
- 4 Adiabatic Flame Temp. can be controlled by excess air



Enthalpy Internal Energy of Combustion

- Enthalpy of combustion h_{RP}
- h_{RP} = Enthalpy of products - Enthalpy of Reactants
- Combustion occurs at a given temperature and pressure
- $h_{RP} = H_P - H_R$
- General form $h_{RP} = \sum_P n_e [h_f^0 + \Delta h]_e - \sum_R n_i [h_f^0 + \Delta h]_i$
- Usual values for fuels (for kg or kmol) given at 25°C and 0.1 MPa
- Internal Energy of Combustion $u_{RP} = U_P - U_R$
- General form

$$u_{RP} = \sum_P n_e [h_f^0 + \Delta h - Pv]_e - \sum_R n_i [h_f^0 + \Delta h - Pv]_i$$
- Considering ideal gas behaviour

$$u_{RP} = h_{RP} - RT(n_{gas-products} - n_{gas-reactants})$$

Second Law Analysis of Reacting Systems

- For steady state steady flow process
- Change in K.E, P.E = 0
- $W_{rev} = \sum n_i(h_i - T_0 s_i) - \sum n_e(h_e - T_0 s_e)$
- General form

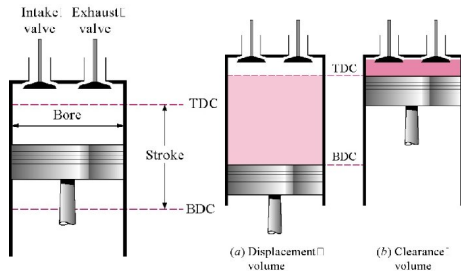
$$W_{rev} = \sum_R n_i [h_f^0 + \Delta h - T_0 s]_i - \sum_P n_e [h_f^0 + \Delta h - T_0 s]_e$$
- Irreversibility $I = \sum_P n_e T_0 s_e - \sum_R n_i T_0 s_i - Q_{cv}$
- Availability of combustion products $\psi = W_{rev} - I$
- For a chemical reaction with reactants and products in temperature equilibrium with surrounding:
- $W_{rev} = \sum_R n_i g_i - \sum_P n_e g_e$, where g = Gibbs function
- Gibbs function at 25 °C and 0.1 MPa is assumed Zero.

Engine Dimensions

Dimensions

- 1 Bore = D
- 2 Stroke = TDC-BDC = L
- 3 Stroke Volume = $\pi L D^2 / 4$
- 4 Clearance Volume = $\pi X D^2 / 4$

Engine Cylinder



Engine piston and crank

Piston



Engine Crank



Air Standard Cycles

Thermodynamic Cycles

- 1 Process
- 2 Cycle
- 3 Representation of the cycle
- 4 First law applied to a process

Assumptions

- 1 Air is the working medium
- 2 Assumes ideal gas behaviour
- 3 Heat addition and removal externally
- 4 Processes are reversible

Nicolas Leonard Sadi Carnot

Father of Thermodynamics

- 1 June 1796 - 24 August 1832
- 2 Was a French military engineer and physicist
- 3 Published - Reflections on the Motive Power of Fire (1824)
- 4 Theorised the maximum efficiency of heat engines.

Nicolas Leonard Sadi Carnot

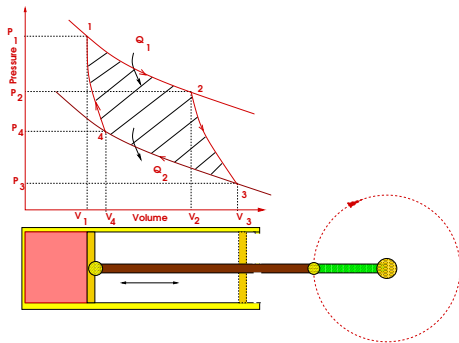


Carnot cycle

Assumptions

- 1 Proposed by Carnot in 1824
- 2 1-2 Rev. Isothermal heat addition
- 3 2-3 Rev. Adiabatic Expansion
- 4 3-4 Rev. Isothermal heat rejection
- 5 4-1 Rev. Adiabatic Compression

Carnot Cycle



Carnot cycle

Analysis

- 1 Heat Supplied rate $= Q_1$,
kJ/s
- 2 Heat Rejected rate $= Q_2$,
kJ/s
- 3 Work $= W = Q_1 - Q_2$
- 4 Efficiency
 $= \eta = [Q_1 - Q_2]/Q_1$.

Carnot Cycle

- 1 maximum efficiency
- 2 Standard for other cycles
- 3 Practical difficulties

Nikolaus Otto

Spark Ignition Engines

- 1 10 June 1832- 26 January 1891, Cologne
- 2 German engineer of the first internal-combustion engine
- 3 The four-stroke engine was first patented by Alphonse Beau de Rochas in 1861

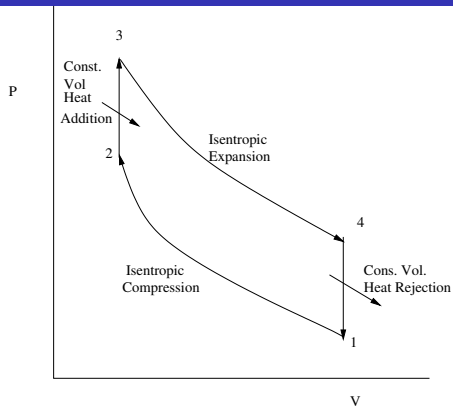
Nikolaus August Otto



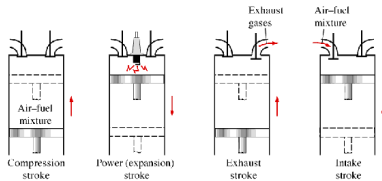
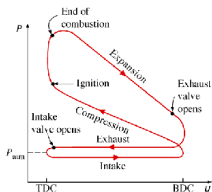
Otto Cycle

Processes

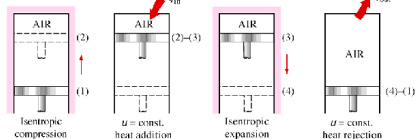
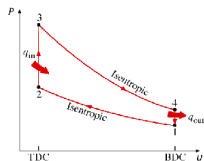
- 1 1-2 Rev. Adiabatic Compression
- 2 2-3 Const. Volume Heat Addition
- 3 3-4 Re. Adiabatic Expansion
- 4 4-1 Const. Volume Heat Rejection



Working of Otto Cycle



(a) Actual four-stroke spark ignition engine



(b) Ideal Otto cycle

a) Actual Otto Cycle working b) Ideal Otto Cycle working

Analysis

$$\text{Heat Supplied} = Q_1 = mc_v(T_3 - T_2)$$

$$\text{Heat Rejected} = Q_2 = mc_v(T_4 - T_1)$$

$$\text{Efficiency} = \eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{(T_4 - T_1)}{(T_3 - T_2)}$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

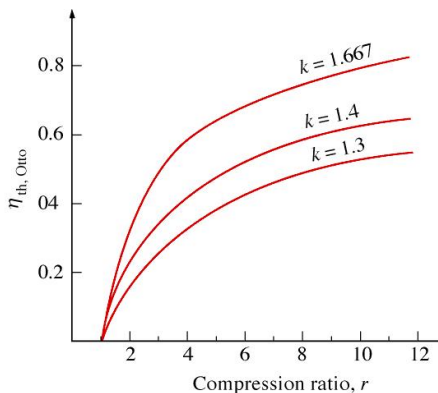
$$\frac{T_3}{T_4} = \left(\frac{v_4}{v_3}\right)^{\gamma-1} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

$$\frac{T_3}{T_2} - 1 = \frac{T_4}{T_1} - 1$$

$$\frac{(T_4 - T_1)}{(T_3 - T_2)} = \frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{\gamma-1}$$

$$\eta_{\text{otto}} = 1 - \left(\frac{v_2}{v_1}\right)^{\gamma-1} = 1 - \frac{1}{r_k^{\gamma-1}}, r_k = \text{Compression ratio} = \frac{v_1}{v_2}$$

Effect of Compression Ratio



- 1 Higher Compression ratio $\Rightarrow$ Higher Efficiency.
- 2 Limited by Engine knock
- 3 γ depends on working fluid

Work Output

$$\text{Work output} = \frac{p_3 V_3 - p_4 V_4}{\gamma - 1} - \frac{p_2 V_2 - p_1 V_1}{\gamma - 1}$$

$$\frac{v_1}{v_2} = \frac{V_1}{V_2} = r_k, V_1 = V_2 r_k = V_4$$

$$\frac{p_2}{p_1} = \frac{p_3}{p_4} = \left[\frac{V_1}{V_2} \right]^\gamma = r_k^\gamma$$

$$\text{let } \frac{p_3}{p_2} = \frac{p_4}{p_1} = r_p$$

$$W_{net} = \frac{p_1 V_1}{\gamma - 1} \left[\frac{p_3 V_3}{p_1 V_1} - \frac{p_4 V_4}{p_1 V_1} - \frac{p_2 V_2}{p_1 V_1} + 1 \right]$$

$$= \frac{p_1 V_1}{\gamma - 1} \left[\frac{r_p r_k^\gamma}{r_k} - r_p - \frac{r_k^\gamma}{r_k} + 1 \right]$$

$$= \frac{p_1 V_1}{\gamma - 1} \left[r_p r_k^{\gamma-1} - r_p - r_k^{\gamma-1} + 1 \right]$$

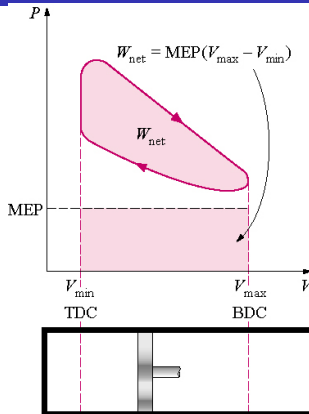
$$W_{net} = \frac{p_1 V_1}{\gamma - 1} (r_p - 1)(r_k^{\gamma-1} - 1)$$

Mean Effective Pressure

- 1 Average pressure with same work out put.
- 2 Work output / Swept volume
- 3 swept
volume = $V_1 - V_2 = V_2(r_k - 1)$

$$4 \quad p_m = \frac{\frac{p_1 V_1}{\gamma - 1} (r_p - 1) (r_k^{\gamma - 1} - 1)}{V_2 (r_k - 1)}$$

$$5 \quad p_m = \frac{p_1 r_k (r_p - 1) (r_k^{\gamma - 1} - 1)}{(\gamma - 1) (r_k - 1)}$$

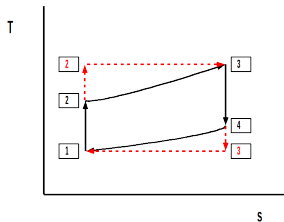


Comparison

- 1 Heat addition Const Temp Vs Const Vol
- 2 Heat rejection Const Temp Vs Const Vol
- 3 Otto efficiency will be less than carnot

Carnot - Otto

Carnot Otto comparison

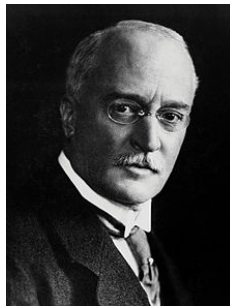


Rudolf Diesel

Compression Ignition Engines

- 1 18 March 1858 to 29 September 1913
- 2 German inventor and mechanical engineer
- 3 Rudolf Diesel obtained patents for his design in Germany and other countries

Rudolf Diesel



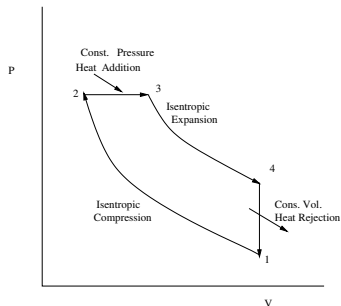
Diesel Cycle

Processes

- 1 1-2 Rev. Adiabatic Compression
- 2 2-3 Const. Pressure Heat Addition
- 3 3-4 Rev. Adiabatic Expansion
- 4 4-1 Const. Volume Heat Rejection

Diesel cycle

P-V Diagram



Analysis

$$\text{Heat Supplied} = Q_1 = mc_p(T_3 - T_2)$$

$$\text{Heat Rejected} = Q_2 = mc_v(T_4 - T_1)$$

$$\text{Efficiency} = \eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{c_v(T_4 - T_1)}{c_p(T_3 - T_2)}$$

$$\eta_{\text{diesel}} = 1 - \frac{(T_4 - T_1)}{\gamma(T_3 - T_2)}$$

$$\text{Compression, } r_k = \frac{V_1}{V_2} = \frac{v_1}{v_2}$$

$$\text{Expansion ratio, } r_e = \frac{V_4}{V_3} = \frac{v_4}{v_3}$$

$$\text{Cut off ratio, } r_c = \frac{V_3}{V_2} = \frac{v_3}{v_2}$$

$$r_k = r_e r_c$$

$$\frac{T_4}{T_3} = \left[\frac{v_3}{v_4} \right]^{\gamma-1} = \frac{1}{r_e^{\gamma-1}}$$

$$T_4 = T_3 \frac{r_c^{\gamma-1}}{r_k^{\gamma-1}}$$

Analysis

Process 2-3, $\frac{T_2}{T_3} = \frac{p_2 v_2}{p_3 v_3} = \frac{v_2}{v_3} = \frac{1}{r_c}$

$$T_2 = T_3 \frac{1}{r_c}$$

Process 1-2, $\frac{T_1}{T_2} = \left[\frac{v_2}{v_1} \right]^{\gamma-1} = \frac{1}{r_k^{\gamma-1}}$

$$T_1 = T_2 \frac{1}{r_k^{\gamma-1}} = \frac{T_3}{r_c r_k^{\gamma-1}}$$

Substituting values of T_1 , T_2 , T_4

$$\eta = 1 - \frac{T_3 \frac{r_c^{\gamma-1}}{r_k^{\gamma-1}} - \frac{T_3}{r_c r_k^{\gamma-1}}}{\gamma \left[T_3 - T_3 \frac{1}{r_c} \right]}$$

$$\eta_{Diesel} = 1 - \frac{r_c^{\gamma} - 1}{\gamma(r_c - 1)r_k^{\gamma-1}}$$

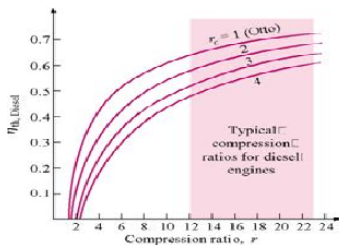
Diesel Compression Ratio

Compression Ratio

- 1 Efficiency increase with compression ratio.
- 2 Efficiency decreases with increase in cut off ratio.
- 3 Cut off ratio depends on the fuel spraying time.

Effect of Compression Ratio

Efficiency Vs Compression Ratio



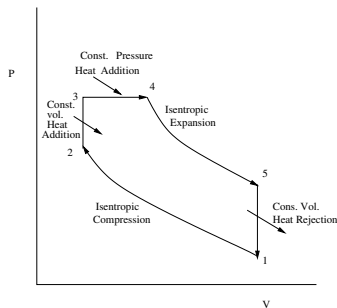
Dual Cycle

Processes

- 1 1-2 Rev. Adiabatic Compression
- 2 2-3 Const. Volume Heat Addition
- 3 3-4 Cons. Pressure Heat Addition
- 4 4-5 Rev. Adiabatic Expansion
- 5 5-1 Const. Volume Heat Rejection

Dual cycle

P-V Diagram



Analysis

Heat supplied $Q_1 = mc_v(T_3 - T_2) + mc_p(T_4 - T_3)$

Heat rejected $Q_2 = mc_v(T_5 - T_1)$

Efficiency $\eta = 1 - \frac{Q_2}{Q_1}$

Compression ratio $r_k = \frac{V_1}{V_2}$

Expansion ratio $r_e = \frac{V_5}{V_4}$

Cut off ratio $r_c = \frac{V_4}{V_3}$

Constant volume pressure ratio $r_p = \frac{P_3}{P_2}$

$$\eta_{Dual} = 1 - \frac{r_p r_c^\gamma - 1}{r_k^{\gamma-1} r_p - 1 + \gamma r_p (r_c - 1)}$$

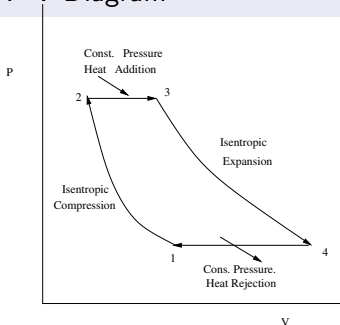
Brayton Cycle

Processes

- 1-2 Rev. Adiabatic Compression
- 2-3 Const. Pressure Heat Addition
- 3-4 Rev. Adiabatic Expansion
- 4-1 Const. Pressure Heat Rejection

Brayton cycle

P-V Diagram



Analysis

$$\text{Heat supplied } Q_1 = mc_p(T_3 - T_2)$$

$$\text{Heat rejected } Q_2 = mc_p(T_4 - T_1)$$

$$\text{Efficiency, } \eta = 1 - \frac{Q_2}{Q_1}$$

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

$$\frac{T_2}{T_1} = \left[\frac{p_2}{p_1} \right]^{\frac{\gamma-1}{\gamma}} = \frac{T_3}{T_4}$$

$$\frac{T_4}{T_1} - 1 = \frac{T_3}{T_2} - 1$$

$$\frac{T_4 - T_1}{T_3 - T_2} = \frac{T_1}{T_2} = \left[\frac{p_1}{p_2} \right]^{\frac{\gamma-1}{\gamma}} = \left[\frac{v_2}{v_1} \right]^{\gamma-1}$$

$$\eta_{\text{Brayton}} = 1 - \left[\frac{v_2}{v_1} \right]^{\gamma-1} = 1 - \frac{1}{r_k^{\gamma-1}} = 1 - \frac{1}{r_p^{\frac{\gamma-1}{\gamma}}}$$

$$r_k = \frac{v_1}{v_2}, r_p = \frac{p_2}{p_1}$$

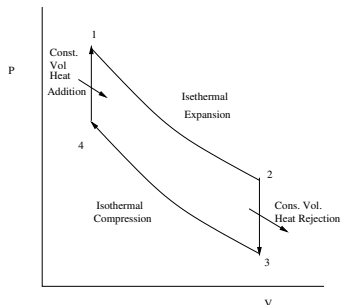
Stirling Cycle

Processes

- 1 1-2 Rev. Isothermal Expansion
- 2 2-3 Const. Volume Heat Rejection
- 3 3-4 Cons. Isothermal Compression
- 4 4-1 Const. Volume Heat Addition

Stirling cycle

P-V Diagram



$$Q_{1-2} = W_{1-2} = RT_1 \ln \frac{v_2}{v_1}$$

$$Q_{2-3} = -c_v(T_2 - T_1)$$

$$Q_{3-4} = W_{3-4} = -RT_2 \ln \frac{v_3}{v_4}$$

$$Q_{4-1} = c_v(T_1 - T_2)$$

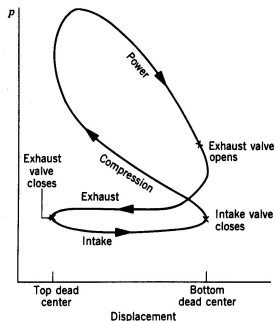
Actual Cycles

Comparison

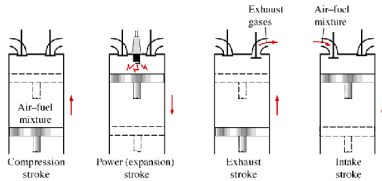
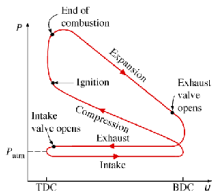
- 1 Working substance, change in chemical composition
- 2 Variation of specific heat
- 3 Dissociation of combustion products
- 4 Progressive combustion, Incomplete combustion
- 5 Heat transfer
- 6 Exhaust blowdown loss
- 7 Gas leakage, friction

Actual cycle

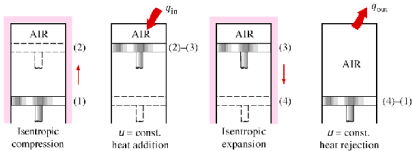
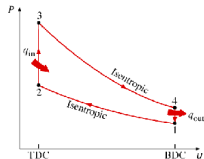
Otto cycle



Actual Working of Otto Cycle



(a) Actual four-stroke spark ignition engine

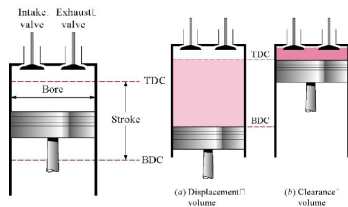


(b) Ideal Otto cycle

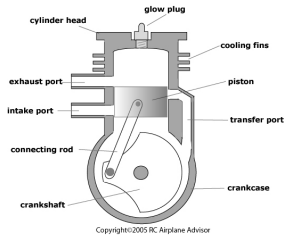
a) Actual Otto Cycle working b) Ideal Otto Cycle working

valves and ports

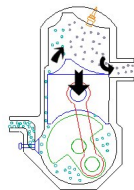
4 stroke engine valves



2 stroke engine

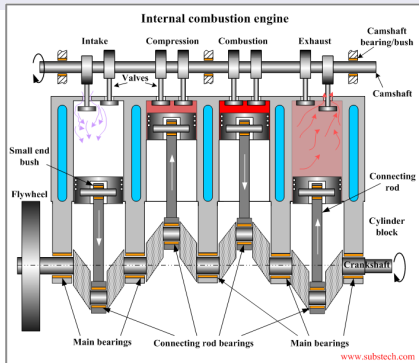


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valves and ports

4 stroke engine valves

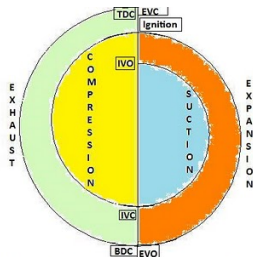


Valves



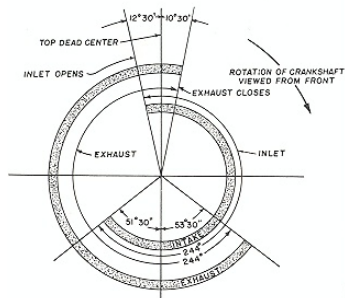
Valve Timing

Theoretical valve timing



Valve Timing Diagram-4stroke engine/otto cycle (Theoretical)

Actual valve timing

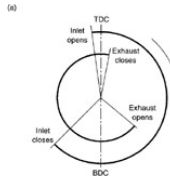


Actual Valve Timing

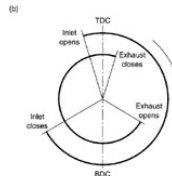
Actual valve timing diagram

Comparison

- 1 Engine Capacity
- 2 Manufacturer Design
- 3 Engine parameters



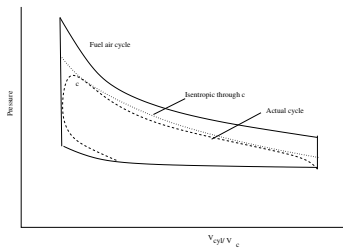
Swept volume 848 cm³
 Compression ratio 8.3:1
 Max. torque ~ 62 Nm at 2900 rev/min
 Max. power ~ 31.3 kW at 4500 rev/min



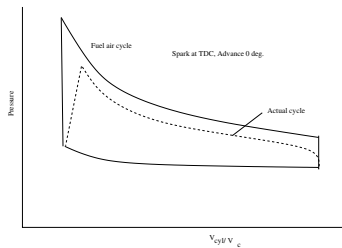
Swept volume 1965 cm³
 Compression ratio 9.2:1
 Max. torque ~ 151 Nm at 3500 rev/min
 Max. power ~ 72 kW at 5200 rev/min

Effect of advancing spark timing

Effect of time loss

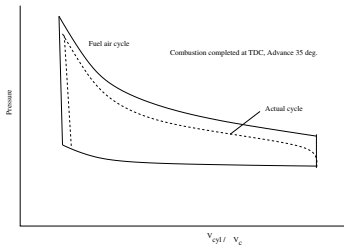


Advance 0°

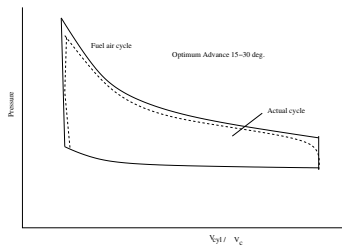


Effect of advancing spark timing

Advance by 35°



Advance optimum $15^\circ - 30^\circ$



Cycle performance for various ignition timings

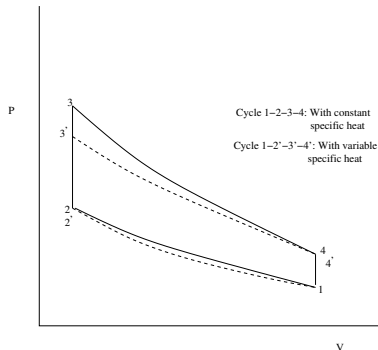
Cycle	Ignition advance	Max cycle pressure, bar	m.e.p bar	Efficiency %	Actual η / Fuel cycle η
Fuel -air cycle	0°	44	10.2	32.2	1.0
Actual	0°	23	7.5	24.1	0.75
	17°	34	8.35	26.3	0.81
	35°	41	7.6	23.9	0.74

c_p variation

Effect of c_p variation

- 1 c_p increases for most of the gases with temperature
- 2 $c_p = 1.005$ kJ/kg at 300K,
 $c_v = 0.717$ kJ/kgK at 300K
- 3 $c_p = 1.343$ kJ/kg at 2000K,
 $c_v = 1.055$ kJ/kgK at 2000K
- 4 γ decreases with increase in temperature
- 5 Loss of work during expansion

Loss of power due to c_p variation



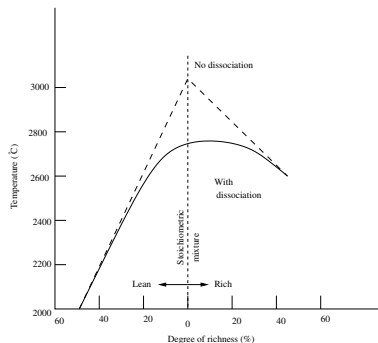
Dissociation

Effect of dissociation

- 1 disintegration of combustion products at high temperature
- 2 at 1000°C ,

$$2\text{CO}_2 + \text{Heat} \rightleftharpoons 2\text{CO} + \text{O}_2$$
- 3 at 1300°C , $\text{H}_2\text{O} + \text{Heat} \rightleftharpoons 2\text{H}_2 + \text{O}_2$
- 4 Dissociation less with rich and lean mixture

Effect of dissociation on temperature

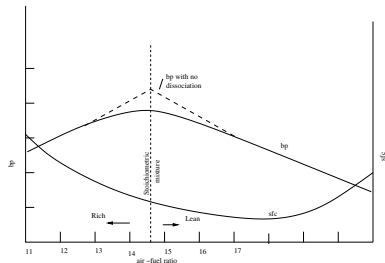


Dissociation

Effect of dissociation on power

- 1 peak temperature reduced
- 2 heat lost through exhaust
- 3 power reduced

Effect of dissociation on temperature



Engine Performance

1 Brake power (kW), $bp = \frac{2\pi NT}{60000}$, $T = (W - s)R$

2 Indicated power (kW) , $ip = \frac{LAnKp_{im}}{60 \times 1000}$

3 Indicated thermal efficiency $\eta_{ith} = \frac{ip}{\text{energy-input}/s}$

$$\eta_{ith} = \frac{ip}{m_f \times C.V}$$

4 Brake thermal efficiency $\eta_{bth} = \frac{bp}{m_f \times C.V}$

5 Mechanical efficiency $\eta_m = \frac{bp}{bp + fp}$

6 Volumetric efficiency

$$\eta_v = \frac{\text{mass-of-charge-actually-inducted}}{\text{mass-of-charge-in-swept-volume-at-ambient}}$$

7 Relative efficiency $\eta_{rel} = \frac{\text{Actual-thermal-efficiency}}{\text{Air-standard-efficiency}}$

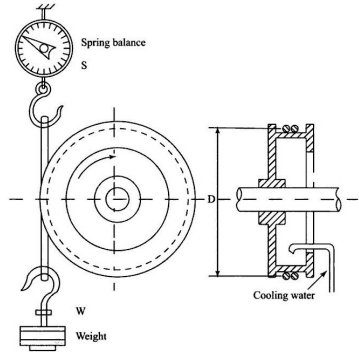
8 Mean effective pressure $p_{im} = \frac{60 \times 1000 \times ip}{LAnK}$

9 specific fuel consumption $sfc = \frac{\text{Fuel-consumption-per-unit-time}}{\text{Power}}$

Engine Testing

Dynamometers

- 1 Rope brake dynamometer
- 2 Electric dynamometer
- 3 Hydraulic dynamometer



Engine Performance

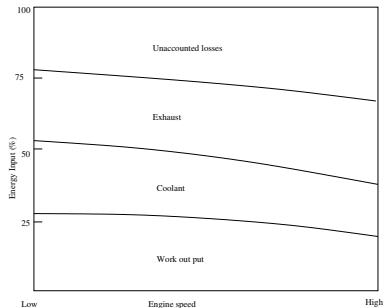
- 1 Air fuel ratio $= \frac{\text{mass-of-air}}{\text{mass-of-fuel}}$
- 2 Equivalence ratio $\phi = \frac{\text{Actual-fuel-air-ratio}}{\text{Stoichiometric-fuel-air-ratio}}$,
 $\phi=1$ - Stoichiometric-fuel-air-ratio,
 $\phi<1$ -lean mixture, $\phi>1$ -rich mixture.
- 3 Mean effective pressure $p_{bm} = \frac{60 \times 1000 \times b p_{ip}}{L A n K}$
 p_{ip} = indicated power, p_{im} = indicated mean effective pressure (N/m^2), L = stroke (m), A = bore area (m^2), N = Engine speed (rpm), n = number of power stroke, = N/2 for 4 stroke and N for 2-stroke, K = number of cylinder

Heat Balance

Distribution

- 1 Work output
- 2 Coolant losses
- 3 Exhaust losses
- 4 Unaccounted losses

Heat balance diagram for a SI Engine

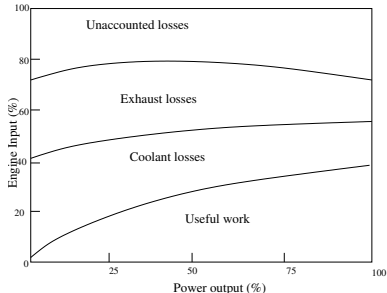


Engine Testing

Performance

- 1 Load test
- 2 Heat balance test
- 3 Morse Test
- 4 Retardation test

Heat balance diagram for a CI Engine



Frictional Power

Morse Test

- 1 Suitable for multiple cylinder engines
- 2 Suitable for SI and CI Engine
- 3 Isolate one cylinder, adjust load to bring rated speed

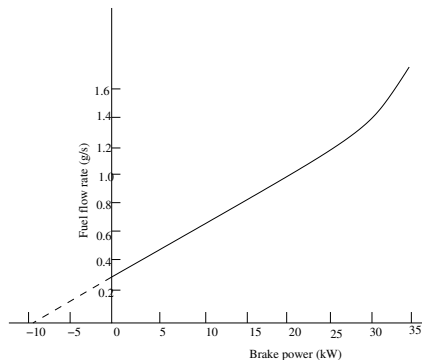
$$4 \quad bp_k = \sum_1^k ip_k - fp$$

$$5 \quad bp_{k-1} = \sum_2^k ip_k - fp$$

$$6 \quad ip_1 = bp_k - bp_{k-1}$$

$$7 \quad ip_k = \sum_1^k ip_k$$

Willan's line method



Retardation test

Test

- 1 Retarding Engine by cutting fuel
- 2 Plot speed vs time
- 3 Repeat test for no-load and 50% load

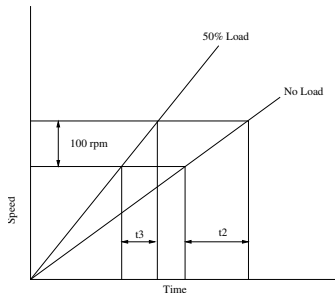
$$4 \quad T = I \frac{d\omega}{dt}, \quad T = MK^2 \frac{d\omega}{dt}$$

$$5 \quad \int_{\omega_1}^{\omega_2} d\omega = \frac{T}{MK^2} \int_{t_1}^{t_2} dt$$

$$6 \quad \text{no load, } \omega_0 - \omega_1 = \frac{T_f t_2}{MK^2}$$

$$7 \quad 50\% \text{ load, } \omega_0 - \omega_1 = \frac{(T_f + T_l) t_3}{MK^2}$$

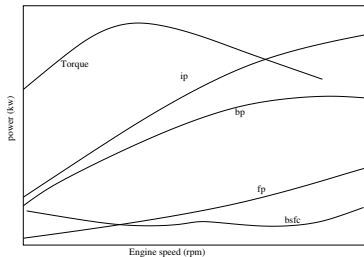
$$8 \quad \frac{t_2}{t_3} = \frac{T_l + T_f}{T_f} \Rightarrow T_f = T_l \left(\frac{t_3}{t_2 - t_3} \right)$$



Engine performance characteristics

- 1 Speed range - maximum charge inducted-maximum torque
- 2 Power output increases with speed
- 3 Air consumption increases with speed
- 4 f_p increases with engine speed
- 5 $bsfc$ drops as speed increased, levels off then increase
- 6 At low speed heat loss high, so poor efficiency, at high speed f_p increases

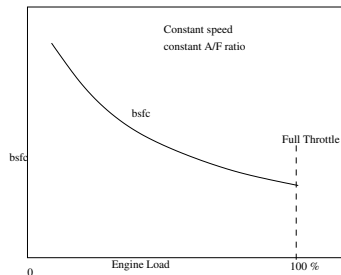
Typical SI Engine performance



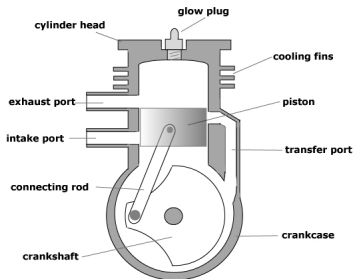
Variables Affecting Engine Performance

- 1 Combustion rate and spark timing, maximum pressure occur beginning of power stroke
- 2 Air fuel ratio to be kept most economic proportions during normal cruising speed
- 3 Higher compression ratio increases thermal efficiency, limited by engine knock
- 4 Low speed $\Rightarrow$ high heat loss, high speed $\Rightarrow$ higher fp
- 5 Mass of inducted charge- maximum possible
- 6 Heat loss - to be minimised

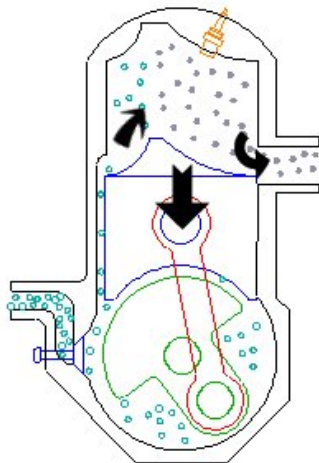
Typical SI Engine performance constant speed



Two stroke engines



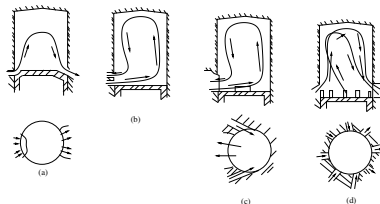
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Scavenging in two stroke engines

- 1 Process of passing compressed charge through transfer port into engine cylinder flushing products of combustion
- 2 Crank case scavenging, separately scavenged engines -using blower or pump
- 3 Cross Scavenging -(a)
- 4 Loop Scavenging, MAN type-(b)
- 5 Loop Scavenging, Schnuerle type (c)
- 6 Loop Scavenging, Curtis type (d)

Methods of Scavenging

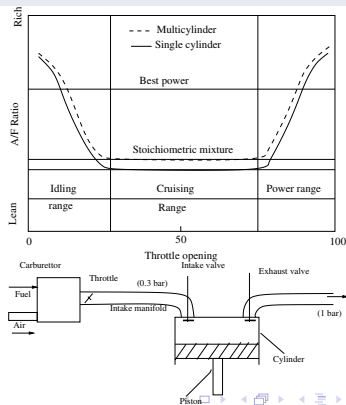


Fuel system

Carburetion

- 1 Process of formation of combustible fuel-air mixture by mixing proper amount of fuel with air
- 2 Require chemically correct mixture, lean mixture, rich mixture
- 3 Idling (Rich mixture), Cruising (lean mixture), High power (Rich mixture)

Carburettor performance and schematic of fuel system

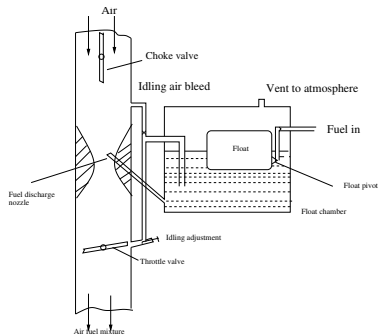


Carburettor

Simple Carburettor

- 1 Float and needle valve maintain level of gasoline
- 2 Pressure decreases at the venturi throat *carburettor depression*
- 3 gasoline engine quantity governed
- 4 simple carburettor fails to maintain A/F ratio

Simple Carburettor



Fuel system CI Engine

Fuel system

- 1 Fuel tank
- 2 Fuel filter
- 3 Fuel pump
- 4 Injector

Typical fuel system for a CI Engine

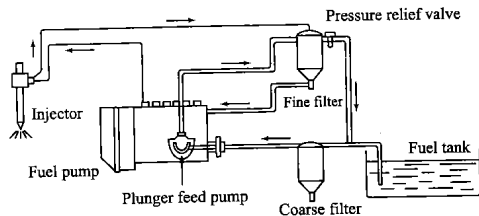


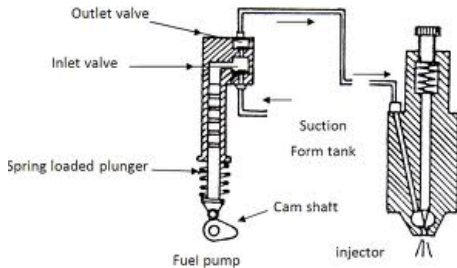
Fig. 9.1 Typical Fuel Feed System for a CI Engine

Fuel feed pump

Fuel feed pump

- 1 At minimum position, plunger creates suction
- 2 Cam turned to maximum lift position plunger goes up
- 3 Fuel delivered through outlet valve

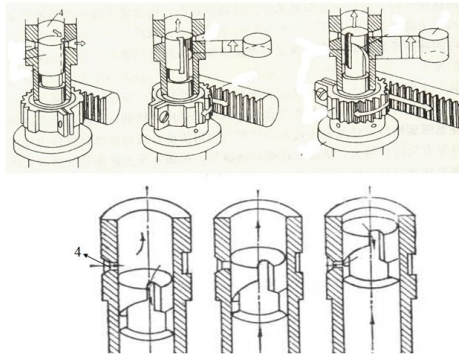
Schematic of fuel feed pump



Injection pump

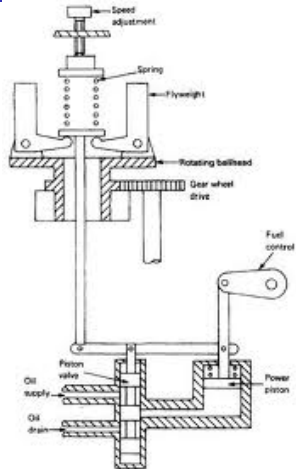
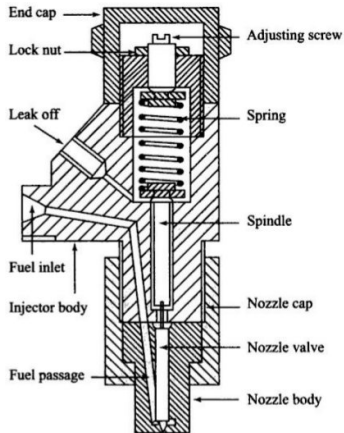
Fuel feed pump

- 1 Pump deliver metered quantity of fuel at high pressure, (120 to 200 x 10^5Pa)
- 2 Axial movement of plunger by Cam, rotation by rack
- 3 Fuel gets filled, as it rises and closes port -fuel injected



Injection pump

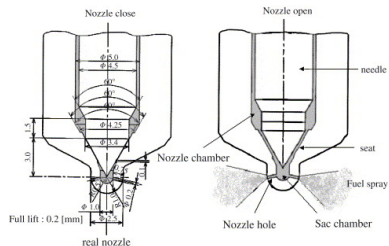
Bosch pump, Mechanical governor



Nozzles

Types of nozzles

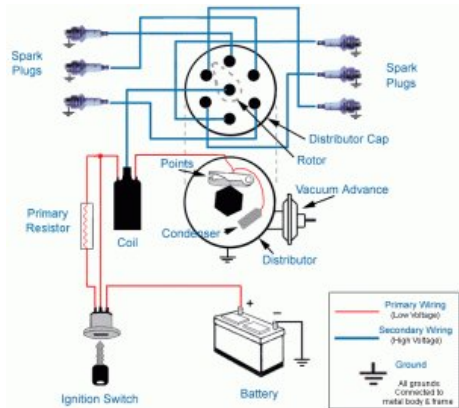
- 1 Pintle nozzles (8 -10 MPa)
- 2 Single hole nozzle (0.2 mm, 8 -10 MPa)
- 3 Multi hole (number 4 to 18, size 35 to 200 μ m, 18 MPa)
- 4 Pintaux Nozzle



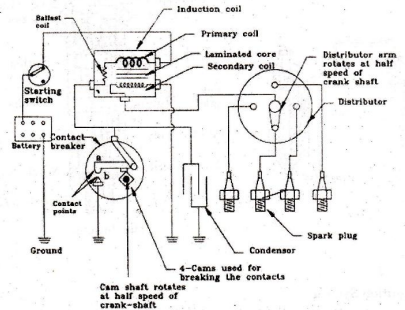
Ignition

Battery Ignition

- 1 battery
- 2 ignition switch
- 3 ignition coil
- 4 contact breaker
- 5 capacitor
- 6 distributor
- 7 spark plug

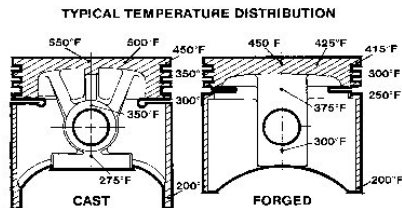


Ignition circuit schematic



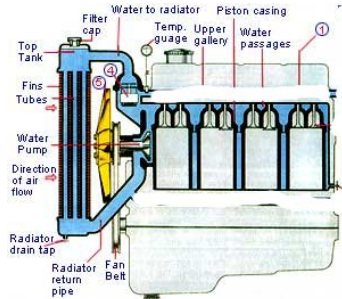
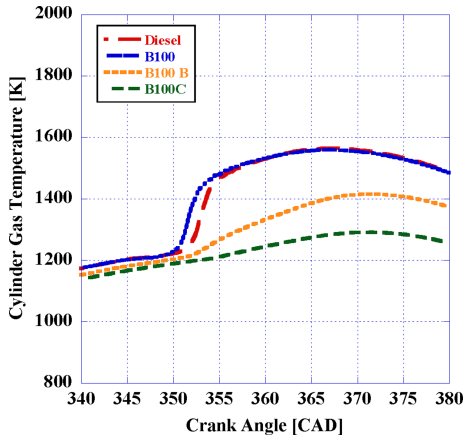
Engine cooling

Temperature variation



Engine cooling circuit

Temperature variation



Lubrication

- 1 Splash lubrication
- 2 Pressurised lubrication

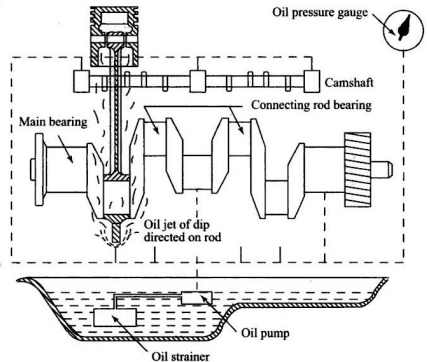
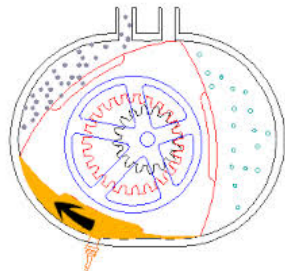


Fig. 13.10 Splash and Pressure Lubrication System

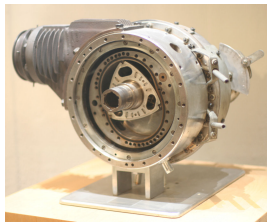
Wankel Engines

- 1 The Wankel engine is a type of internal combustion engine using an eccentric rotary design to convert pressure into rotating motion.
- 2 The engine was invented by German engineer Felix Wankel. He received his first patent for the engine in 1929.
- 3 Intake, compression, Ignition, Expansion, Exhaust.
- 4 Each combustion chamber in the Wankel engine generates one combustion stroke per driveshaft rotation. i.e, three power strokes per rotor rotation.



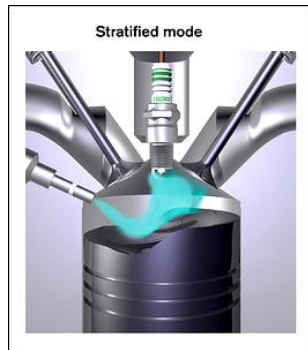
Wankel Engines

- 1 The Prime advantages of the Wankel engine are:
A far higher power to weight ratio than a piston engine. No reciprocating parts. Runs with almost no vibration. Not prone to engine-knock. Cheaper to mass-produce as contains few parts.
- 2 Disadvantages : Rotor sealing - problem - different temperatures in each separate chamber section. The combustion is slow as the combustion chamber is big and moving. Poor fuel consumption - exhaust stream is enriched with unburned mixture and carbon monoxide. Poor emissions.



Stratified charge engine

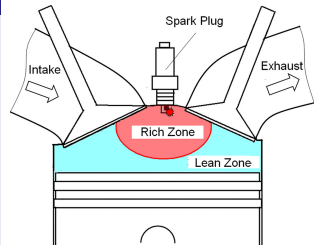
- 1 The fuel is injected into the cylinder just before ignition
- 2 This allows for higher compression ratios without "knock,"
- 3 Intake, compression, Ignition, Expansion, Exhaust.
- 4 Leaner air/fuel ratio than in conventional internal combustion engines.



Stratified charge engine

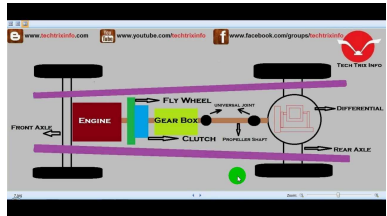
Disadvantages

- 1 Increased injector cost and complexity
- 2 Higher fuel pressure requirements
- 3 Carbon build-up on the back of the intake valve due to the lack of gasoline passing by the intake valve to act as a cleaning agent for the valve on traditional multi-port injection designs



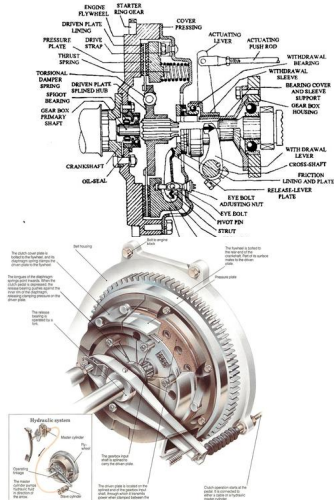
Transmission

- 1 A machine consists of a power source and a power transmission system
- 2 which provides controlled application of the power
- 3 Often transmission refers simply to the gearbox that uses gears and gear trains to provide speed and torque conversions from a rotating power source to another device



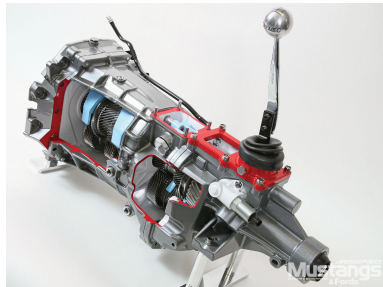
Clutch

- 1 A clutch is a mechanical device that engages and disengages the power transmission.
- 2 In the simplest application, clutches connect and disconnect two rotating shafts (drive shafts or line shafts)
- 3 Friction clutches, Centrifugal, Cone clutch



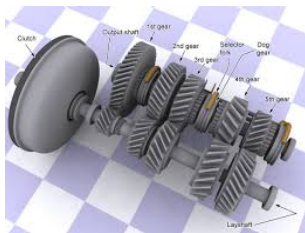
Gear Box

- 1 Most modern gearboxes are used to increase torque while reducing the speed of a prime mover output shaft (e.g. a motor crankshaft)
- 2 Engines typically operate over a range of 600 to about 7000 revolutions per minute
- 3 the car's wheels rotate between 0 rpm and around 1800 rpm.
- 4 Manual transmissions can include both synchronized and unsynchronized gearing



Gear Box

- 1 A simple but rugged sliding-mesh or unsynchronized/non-synchronous system, where straight-cut spur gear sets spin freely.
- 2 Synchronized by the operator matching engine revs to road speed, to avoid noisy and damaging clashing of the gears
- 3 Some commercial applications use non-synchronized manual transmissions that require a skilled operator
- 4 Automatic transmission that selects an appropriate gear ratio without any operator intervention



Differential

- 1 A differential couples the input shaft (or prop shaft) to the Pinion,
- 2 The differential gearing allows the outer drive wheel to rotate faster than the inner drive wheel during a turn.
- 3 A differential consists of one input, the drive shaft, and two outputs which are the two drive wheels

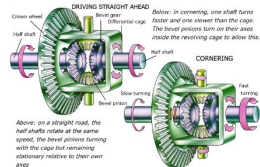
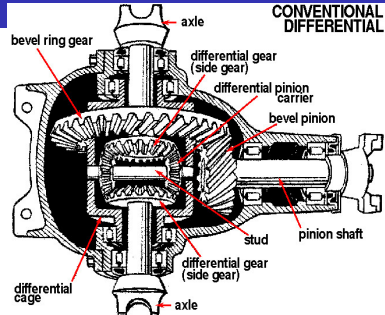
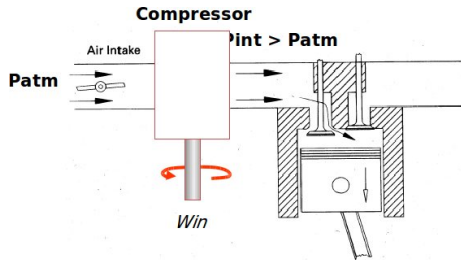


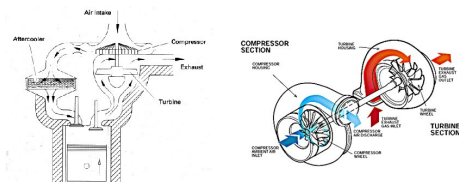
Fig 1. Differential - how it works

Supercharger

- 1 These devices are used to increase the power of an IC engine by raising the intake pressure and thus allowing more fuel to be burned per cycle.
- 2 Superchargers are compressors that are mechanically driven by the engine crankshaft and thus represents a parasitic load.



Turbocharger



- 1 Turbochargers couple a compressor with a turbine driven by the exhaust gas. The compressor pressure is proportional to the engine speed
- 2 Compressor also raises the gas temperature, so after-coolers are used after the compressor to drop the temperature and thus increase the air density.

Exhaust emissions

- 1 Oxides of nitrogen - Engine design, operating conditions, maximum at 14:1 A/F ratio, low at lean and rich mixture
- 2 Carbon monoxide - CO present in lean mixtures, high at idling, deceleration
- 3 unburnt hydrocarbons - incomplete combustion
- 4 Aldehydes Odourous oxygenated hydrocarbons - generally carcinogenic
- 5 Smoke

Emissions

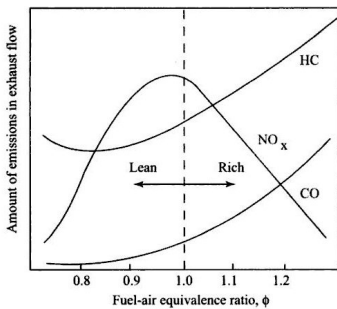


Fig. 15.1 Emissions as a Function of Equivalence Ratio for an SI Engine

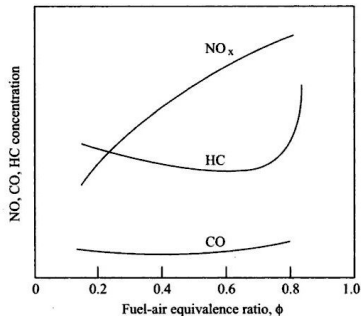
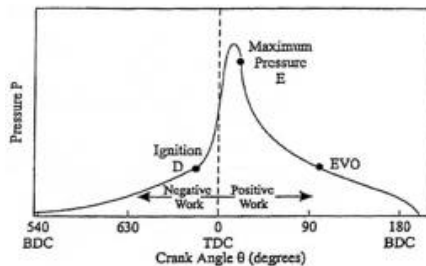
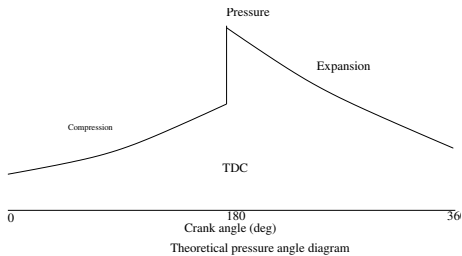


Fig. 15.2 Emissions as a Function of Equivalence Ratio for a CI Engine

Combustion



Combustion in SI Engine

Combustion in SI Engine

- 1 Ignition lag-preparation phase, Chemical process depends on temp and pressure.
- 2 Propagation of flame
-Spread of flame,
measurable rise of pressure.
- 3 After burning -starts with maximum pressure point, flame velocity decreases
Some additives are harmful

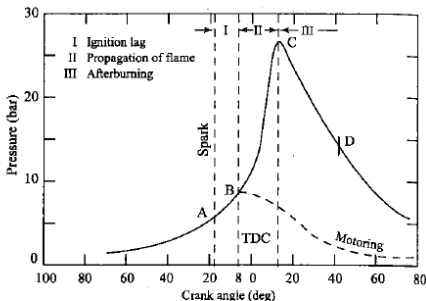


Fig. 12.2 Stages of Combustion in an SI Engine

HUCR, Fuel Rating

- HUCR-Highest Useful Compression Ratio (HUCR) is the compression ratio at which a fuel can be used without detonation.
- at HUCR engine under specified operating conditions, give best efficiency.
- Fuel Rating - S.I Engine fuels - Octane number
- Iso Octane (C_8H_{18}) good anti-knock quality assigned 100 Octane number
- Normal Heptane(C_7H_{16}) poor anti-knock quality assigned 0 Octane number
- Sample fuel compared with % by volume of mixture matching exact knock intensity

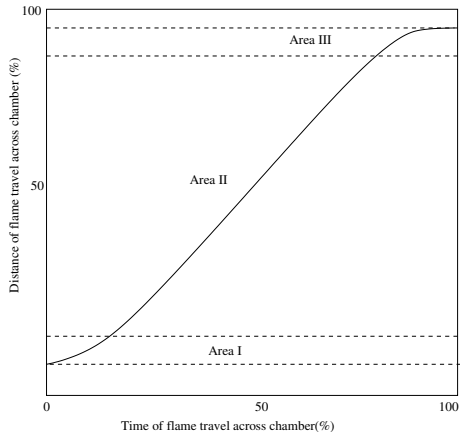
Fuel Rating, additives

- Fuel Rating - C.I Engine fuels - Cetane Number
- Cetane ($C_{16}H_{34}$) good anti-knock quality assigned 100 cetane number
- Alpha methyl naphthalene ($C_{11}H_{10}$) poor anti-knock quality assigned 0 cetane number
- Sample fuel compared with % by volume of mixture matching exact knock intensity
- Gasoline additives increase gasoline's octane rating (ex: tetra ethyl lead) or act as corrosion inhibitors or lubricants.
- Allowing the use of higher compression ratios for greater efficiency and power.
- Types of additives include metal deactivators, corrosion inhibitors, oxygenates and antioxidants.

Flame front propagation

Efficient combustion

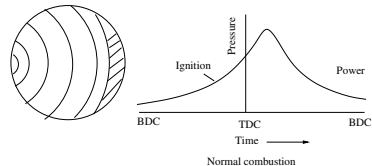
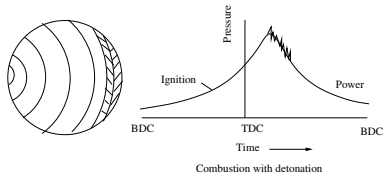
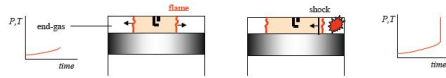
- 1 Propagation of flame front critical
- 2 reaction rate - chemical combination process
- 3 transposition rate - physical movement of the flame front



Abnormal combustion

Knock in SI Engines

- 1 Temperature of unburnt mixture exceeds self-ignition temperature during preflame reaction
- 2 Autoignition occurs at various pin-point locations - knocking
- 3 Opposing flame front may collide creating pressure pulse
- 4 depends on fuel properties
- 5 high autoignition temperature and a long ignition delay desirable for SI engine



Combustion Chambers for SI Engines

Combustion Chambers for SI Engines

- 1 Density factors - Compression ratio, Mass of charge, Inlet temperature, Temp of walls
- 2 Time factors - Turbulence, Engine speed, Flame travel distance, Engine size, Combustion chamber shape, Spark plug location
- 3 Composition factors- Fuel Air ratio, Octane Value of fuel. Octane number 100 for iso octane (C_8H_{18}), 0 for normal heptane (C_7H_{16})

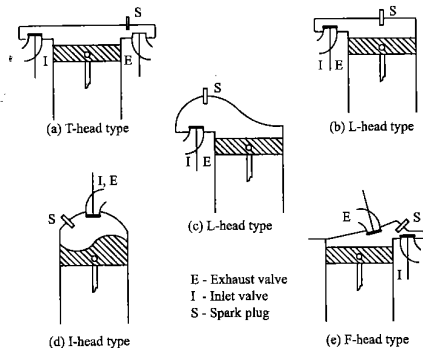


Fig. 12.8 Examples of Typical Combustion Chamber

Combustion in CI Engines

- 1 Ignition delay -period of inactivity after the first droplet hit the hot air and actual combustion
- 2 Physical delay - time between beginning of injection and attainment of chemical reaction
- 3 Chemical delay - Reaction start slowly and then accelerate

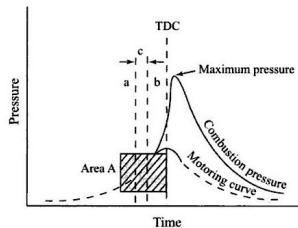
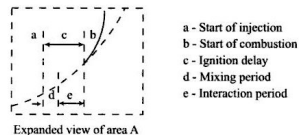


Fig. 12.12 Pressure-Time Diagram illustrating Ignition Delay

Factors affecting combustion delay

- 1 Design and operating factors- compression ratio, engine speed, injection timing, quality of the fuel -cetane number reduce self ignition temperature
- 2 Cetane number 100 for normal cetane ($C_{16}H_{34}$), 0 for alpha methyl naphthalene ($C_{11}H_{10}$)
- 3 Knock in CI Engine - results from longer ignition delay

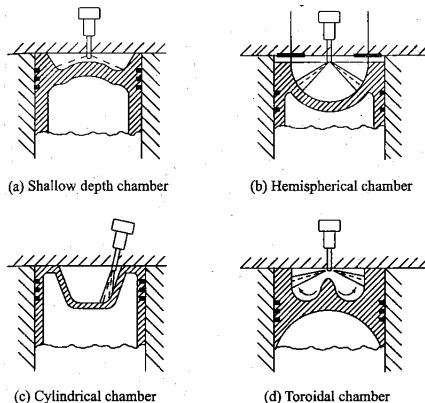


Fig. 12.19 Open Combustion Chambers